Lab 3 – Extra Questions

Part B: Programming Exercises

1. Write and test the program that allows the user to find the area of a rectangle or the area of a circle by prompting the user to enter the character 'R' or 'r' for rectangle and 'C' or 'c' for circle into a variable named choice. Base on the user input, the program will then ask user to inter the dimensions that are required to calculate the area. The area of a rectangle is (length * width) and the area of a circle is (π * radius^2) where π is 3.14159. Define global constant for π and use the pow function in your calculation. Use a loop to allow user to calculate 5 different areas.

   Note: Use **if statements** to decide between circle or rectangle.

2. Repeat question 1 above by using **switch statement**.

3. Write a program that asks a user to enter two characters into two variables named ch1 and ch2. The program checks if ch2 is lower than ch1 in the alphabetical ordering. If yes, the program swaps the characters and print out the ch1 and ch2 before and after swapping. Otherwise, each character ch1 and ch2 will be increased by 2 and the program displays the characters ch1 and ch2 before and after the increment.

   SWAP 2 character:

   ```
   temp = ch1;
   ch1 = ch2;
   ch2 = temp;
   ```

   For example, if ch1 = 'k' and ch2 = 'e', then ch2 is lower than ch1, then the program will swap the two characters and the output should look like the following:

   Before swapping: ch1 = 'k' and ch2 = 'e'
   After swapping: ch1 = 'e' and ch2 = 'k'

   For example, if ch1 = 'b' and ch2 = 'w', then ch2 is not lower than ch1, then the program will increase the two characters by 2 and the output should look like the following:

   Before increment: ch1 = 'b' and ch2 = 'w'
   After increment: ch1 = 'd' and ch2 = 'y'

   Note: Use a while-loop to repeat the above. The while-loop will terminate when the user decides not to continue.

4. Write a C++ program that reads in assignment mark, mid-term exam mark and final exam mark of a student. Then calculate the average mark by adding the 3 marks and divide by 3. If the average mark is less than 50, output fail; otherwise, output pass. The sample input and output of the program is as follow:

```
Enter the assignment mark:  75
Enter the mid-term exam mark:  80
Enter the assignment mark:  60

The student's average mark is 71.7
The student has passed.
```

```
Enter the assignment mark:  70
Enter the mid-term exam mark:  10
Enter the assignment mark:  30

The student's average mark is 36.7
The student has failed.
```

   Use a loop to repeat the above process for 5 different students.

5. Write a C++ program that reads in two whole numbers and outputs their sum and average. Be sure to prompt for input, echo input, and label all output. Use a loop to input and output the results as shown below. Note: The loop will terminate when one of two integers entered by the user is a negative integer.

```
1.  Enter 2 integers:  20   30
    The sum = 50 and the average = 25.00

2.  Enter 2 integers:  100   25
    The sum = 125 and the average = 62.50

3.  Enter 2 integers:  16   45
    The sum = 61 and the average = 30.50

4.  Enter 2 integers:  38   82
    The sum = 120 and the average = 60.00

5.  Enter 2 integers:  71   30
    The sum = 101 and the average = 50.50

6.  Enter 2 integers:  58   -30
    End of the program.

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```